01

The Journey from Data to Decisions

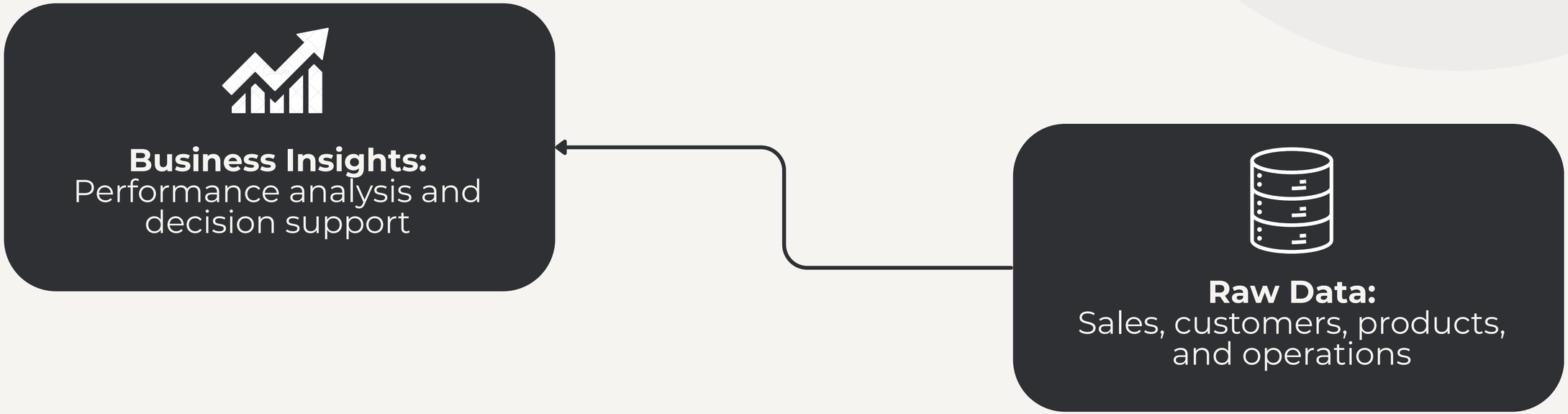
RETAIL SALES PERFORMANCE ANALYTICS

Presenters:

Aya Hisham, Rayan Mohamed, Ahmed Maher, Aya Anwar, Shrouk Soliman

THE PROJECT AT A GLANCE

Turning retail transaction data into actionable business insights.



Our Focus

Sales Performance · Customer Insights · Operational Analysis · Regional Performance

OUR JOURNEY

From Data to Decisions

Phase 1 **Data Cleaning:** Preparing reliable and consistent data.

↓

Phase 2 **Data Modeling:** Structuring data and defining relationships.

↓

Phase 3 **DAX Measures:** Calculating meaningful performance metrics.

↓

Phase 4 **Dashboard & Insights:** Visualizing results and supporting decisions.

Phase 1

04

DATA CLEANING

From Raw Data to Reliable Insights



Collect & Combine:
Automated append of yearly files via Folder Connector.



Detect Issues:
Identified nulls, duplicates, and invalid outliers.



Clean & Standardize:
Corrected data types, trimmed text, and validated ranges.

Phase 2

DATA MODELING

05

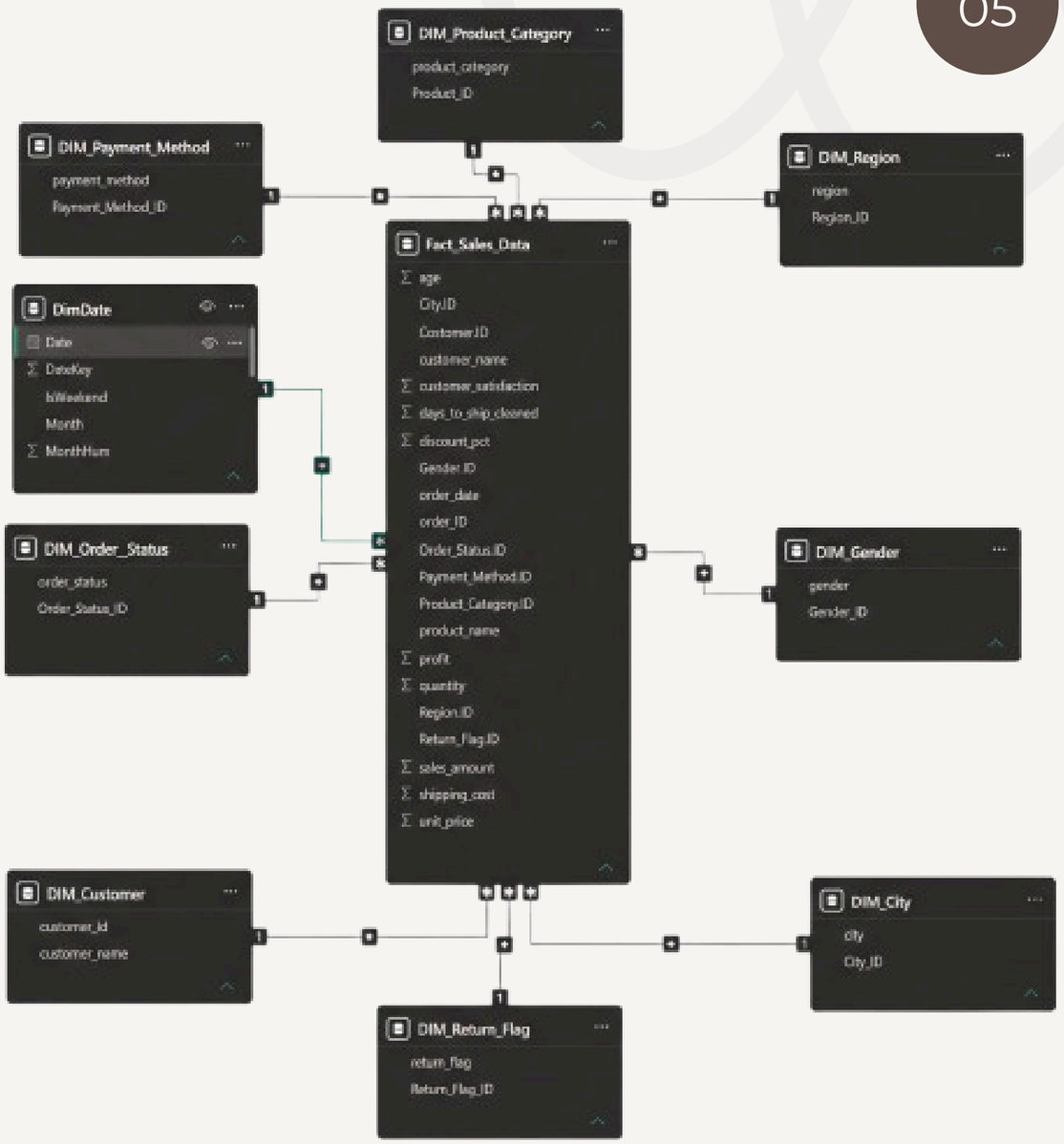
Turning Clean Data into a Well-Designed Structure

01 The House Analogy

02 Star Schema

03 One-to-Many Relationships

04 Analysis Foundation



Phase 3

ADVANCED DAX ENGINE

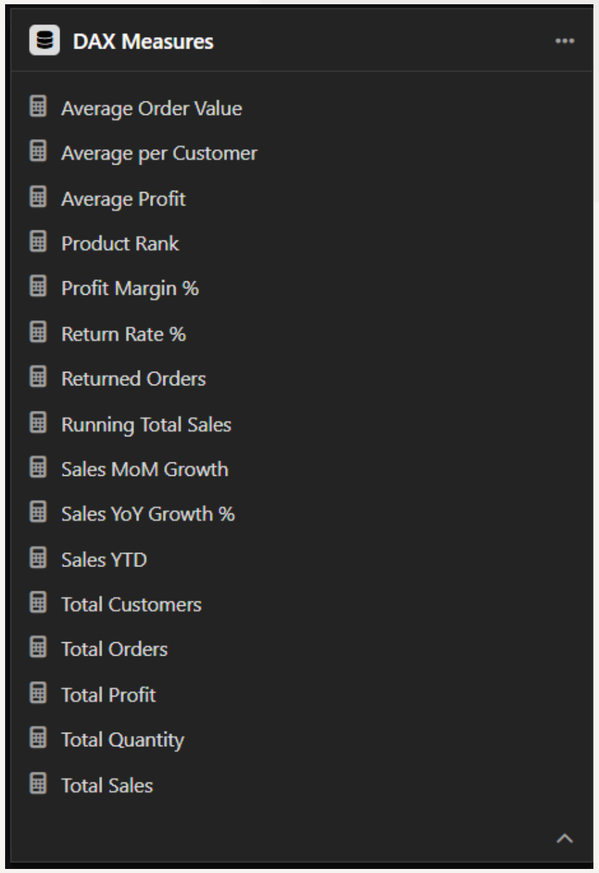
06

Powering the dashboard with dynamic business logic and complex calculations.

Core Business Metrics:
Calculated Total Sales (SUM) & Profit Margin (DIVIDE) to evaluate overall performance and profitability.

Advanced Time Intelligence:
Utilized CALCULATE, FILTER, & ALL for Running Total Sales to track cumulative growth over time.

Dynamic Ranking & Insights:
Implemented RANKX for Product Rank to instantly identify top-selling products.

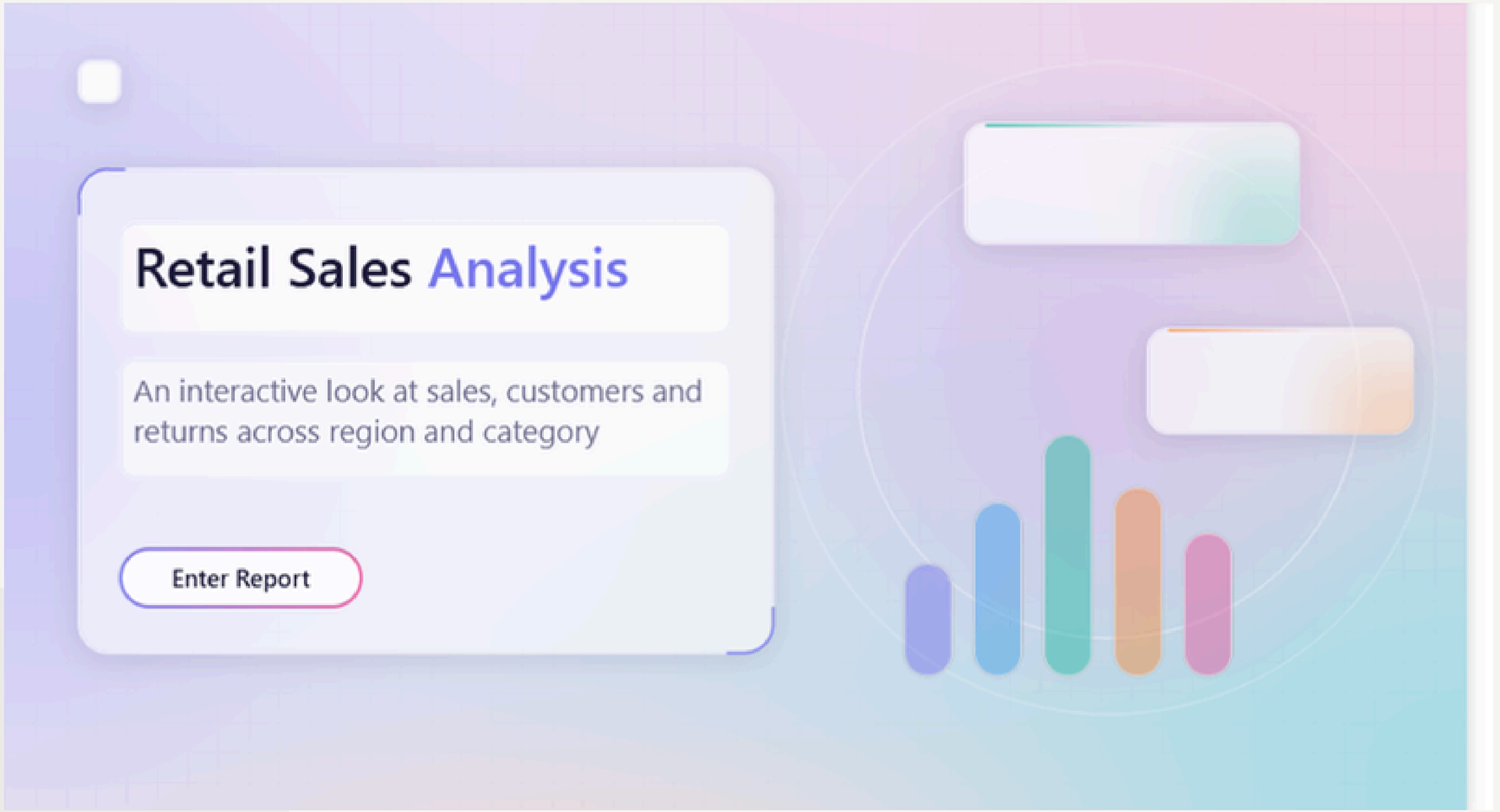


Phase 4

07

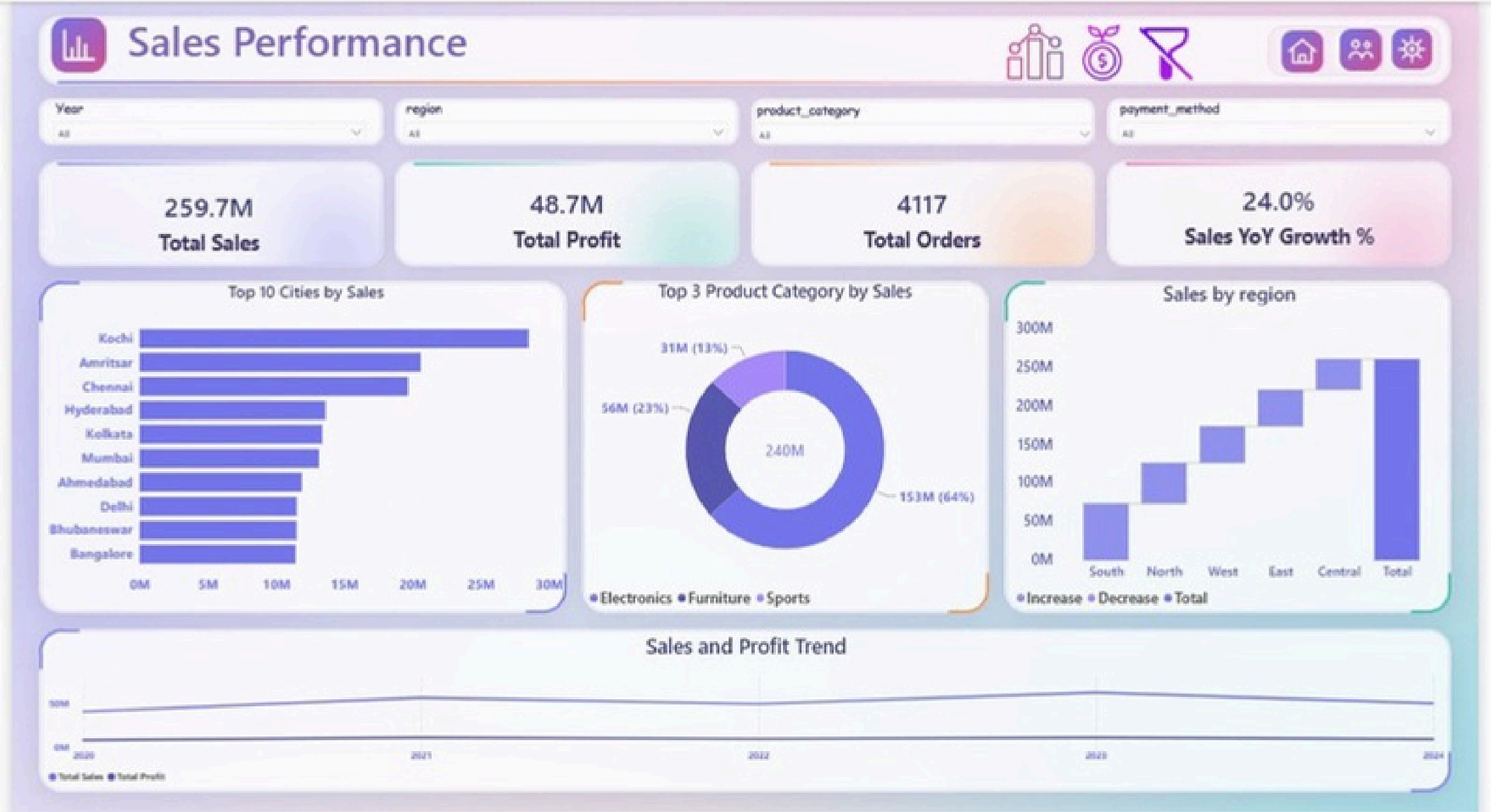
FROM ANALYSIS TO VISUALIZATION

An interactive dashboard designed to turn data into actionable insights.



SALES PERFORMANCE

How is the business performing in terms of sales and profitability?



REGIONAL PERFORMANCE

How does performance vary across regions and countries?



city	product_category	Total Orders	Returned Orders	Return Rate %
Bhubaneswar	Beauty	2	1	50.00%
Bhubaneswar	Books	1	1	100.00%
Bhubaneswar	Clothing	1		
Bhubaneswar	Electronics	4		
Bhubaneswar	Furniture	1		
Guwahati	Beauty	2		
Guwahati	Clothing	1		
Guwahati	Electronics	1		
Guwahati	Furniture	1		

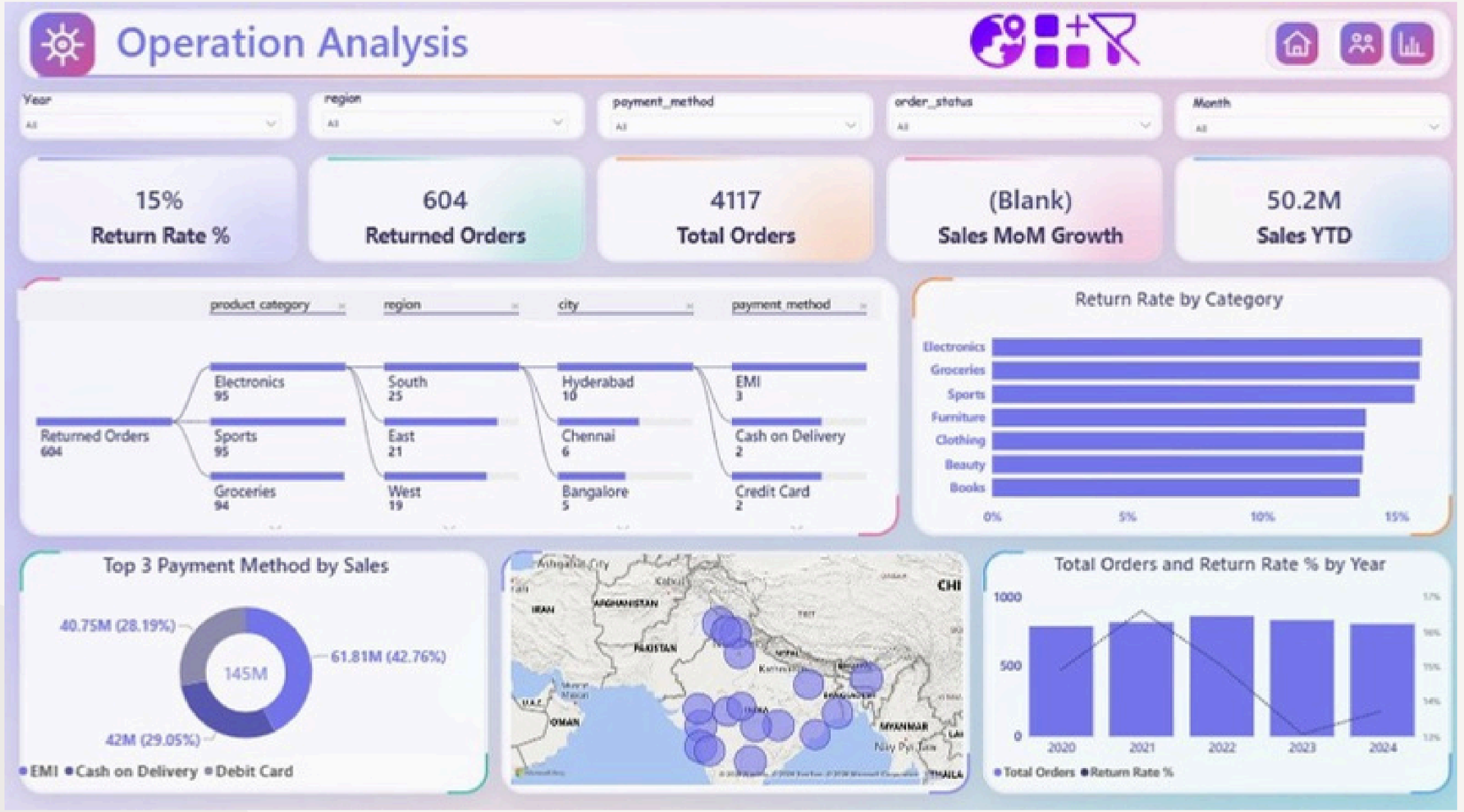
CUSTOMER INSIGHTS

What can customer data tell us about purchasing behavior?



OPERATIONAL ANALYSIS

What operational patterns can we identify?



FROM DATA TO DECISIONS

How every stage contributes to the final outcome

Reliable Data



DATA CLEANING:
Improving data quality and consistency.

Structured Analysis



DATA MODELING + DAX:
Organizing data and calculating meaningful metrics.

Actionable Insights



DASHBOARD:
Exploring performance and supporting business decisions.

One connected process. One clearer view of business performance.

ROW-LEVEL SECURITY (RLS)

Restricting data visibility based on user permissions.

Controlled Access:
Users can access only the rows permitted by their assigned roles.

Role-Based Filtering:
Data visibility is determined by the configured security rules.

Secure Reporting:
Helps prevent unauthorized access to restricted data rows.

THANK YOU

Numbers tell A story, but insights
shape the future.

Opening the floor for questions and discussion